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A vehicle routing algorithm for the distribution of fresh vegetables and similar perishable food

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Abstract

An algorithm for the distribution of fresh vegetables in which the perishability represents a critical factor was developed. This particular problem was formulated as a vehicle routing problem with time windows and time-dependent travel-times (VRPTWTD) where the travel-times between two locations depends on both the distance and on the time of the day. The model considers the impact of the perishability as part of the overall distribution costs and a heuristic approach, based on the tabu search is used to solve the problem. The performance of the algorithm was verified using modified Solomon's problems. Using parameters typical of the Slovenian food market, different schedules were achieved, giving improvements of up to 47% reduction in perished goods.
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Keywords: Vehicle routing problem; Distribution; Perishable food; Time-dependent; Loss of quality

1. Introduction

The difficulty in preserving the nutritional characteristics of fresh food-stuffs during transportation presents a direct problem to food distributors where the perishability of the produce requires it to be handled in ways not necessarily conducive to the traditional view of cost effective distribution activities. Fresh vegetables provide a representative example of perishable goods; the nutrition value and taste are at their best directly after harvesting, decreasing as time elapses until the food is spoilt.

In the Slovenian market, a quick distribution of fresh vegetables is required since, of the approximately 151,571,000 kg of fresh vegetables consumed per year, between 50% and 70% are imported, mainly from Italy and Spain. When they arrive in Slovenia, they are first delivered to the distribution centers and then distributed to the final consumers who can acquire the product in optimal conditions only 48 h after the harvesting. In the transportation of vegetables insulated vehicles are used for international transportation and non-insulated vehicles for the final distribution. A detailed description of the vegetables consumed, the typical quantities and the distribution process are presented in Vadnal and Bratuša (2002), Križaj (2002) and Knavs (2003).

To measure the decrease in the value of a load of fresh vegetables, we define the "quality" of the load. The quality is 100% when the load can be sold entirely at the current market price and the quality is 0% when the load has lost its commercial value. The loss of quality in the transportation of the vegetables represents a significant cost for the companies. By Slovenian law, the recognized value of the loss is 8%, but occasionally the loss can be higher and have been reported by transportation companies as up to 23% (Križaj, 2002).

Goods-distribution companies try to reduce basic distribution costs based on the number of vehicles used, total distance-traveled and the total travel-time. The standard view is that in addressing basic distribution issues distributors also

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Notations

N_i number of the customers for vehicle i
C_{ji} load transported by the vehicle i on the path between the customer j and the customer $j+1$
Travel-time$_{j(j+1)}$ time spent by vehicle i between the start of servicing of customer j and the start of the servicing of customer $(j+1)$
S_i^k arrival time of vehicle k at customer i
DelayedCostFactor$_i$ cost for a minute of lateness defined for customer i
NV number of vehicles
Distance$_i$ distance in km traveled by the vehicle i
Time$_i$ time in minutes that the vehicle i spend on the route
Penalty$_i$ time of delay in minutes that the vehicle i accumulate on the route and calculated as in (2)
Load×Time$_i$ loss of quality at vehicle i, calculated as in (1)
k_1, k_2, k_3 and k_4 parameters that represent the weight of each part of the total cost.

indirectly reduce the loss of quality. In this paper, we present a model for the representation of the loss of quality and consider it as part of the overall distribution costs. We model the distribution problem between the distribution centers and the customers (retailers) as a vehicle routing problem with time windows and with time-dependent travel-times (VRPTWTD) where the travel-times between two locations depends on both the distance and on the time of day. To minimize the overall distribution cost, the objective function must not only model the number of vehicles, the total distance-traveled and the total travel-time, but additionally the loss of quality of the load.

2. Literature review

Different methods have been developed to solve the vehicle routing problem (VRP) and vehicle routing problem with time windows (VRPTW). The routes should be chosen to minimize the total distribution cost. A detailed description of these and other related problems including a literature review of the methods are provided in Toth and Vigo (2002). In the VRPTW we know the location, the demand and the delivery window (the earliest and latest time that the customer is able to accept the load) of each customer. The solution is presented by one or more routes, each associated to one vehicle (a route begins at the depot, traverses a subset of customers in a specified sequence and returns to the depot). Each customer must be assigned only to one vehicle and the load must not exceed the vehicle capacity.

Various works address the transportation of foodstuffs, dealing with pertinent issues to our problem. Tarantilis and Kiranoudis (2001) analyzed the distribution of the fresh milk. They formulated the problem as a heterogenous fixed fleet vehicle routing problem; this is a VRP with vehicles that have different capacity. A threshold-accepting based algorithm was developed aiming to satisfy the distribution needs of the company, allowing them to schedule their distribution many times a week. Tarantilis and Kiranoudis (2002) presented a real-life distribution problem of fresh meat in an area of the city of Athens. They formulated the problem as an open multi-depot vehicle routing problem. They presented a new stochastic search meta-heuristic algorithm belonging to the class of threshold-accepting algorithms. Hwang (1999) presented an effective distribution model for determining optimal patterns of food supply and inventory allocation for famine relief areas. He modeled a VRP that incorporated inventory allocation and optimal distribution based on minimizing the deprivation and starving instead of travel distance or time. Prindezis, Kiranoudis, and Marinos-Kouris (2003) presented an application service provider, to be used for central food markets, which coordinates and disseminates tasks and related information for solving the VRP. For the solution of the VRP they used a meta-heuristic technique based on the tabu search. They tailored their software to the road network of Athens and applied it to the integrated-logistics problem of deliveries to the 690 retail companies that comprise the Athens Central Food Market. They used a two-phase algorithm to solve the VRP. In the first phase, a route construction algorithm was used and in the second phase a tabu search was used to improve the given solution. Faulin (2003) presented the implementation of the mixed algorithm procedure that uses heuristic and exact subroutines in the solution of a VRP having specific constraints related to companies in the agribusiness field.

In all the above models, the travel-time between the customers was constant and none of them took into account the specific degradation of quality during transport.

Amponsah and Salhi (2004) presented an efficient heuristic for the routing problem of the collection of garbage. They considered it as a capacitated arc routing problem (CARP) and minimize the environmental effect due to the smell of the garbage and total collection cost in a bi-objective model. In CARP the customer (garbage to be collected) is associated to the arcs and not to the nodes as in the VRP. The smell of the garbage gets worse with time and so it is necessary to remove it as early as possible,

which is analogous to food deterioration. They defined inconvenience due to smell by the total quantity of garbage over time. They used a look-ahead strategy that gives the possibility to choose from a pool of solutions the one which best solves the problem.

In most real-world distribution problems, it is important to consider the fluctuations of the travel-time in the solution of the problem. Travel-time plays an important role in the distribution of the perishable goods, since its fluctuations may extend the time that the goods spend on the vehicles. Different representations of the fluctuations of the travel-times between the customers have been reported and different extensions of VRP have been proposed to address the fluctuations in travel-times. Routing problems with stochastic travel-times are presented by Gendreau, Laporte, and Seguin (1996), Fu (2002), Hadjiconstantinou and Roberts (1998) and Kenyon and Morton (2003).

The routing problem that includes time of day-dependent travel-time was first introduced by Malandraki and Daskin (1992). They considered the time-dependent traveling salesman problem (TSP), that is a simplified VRP problem in which all customers must be visited by only one vehicle. Bentner, Bauer, Obermair, and Morgenstern (2001) presented a TSP in which they have considered a zone in the city center with traffic jams in the afternoon and show how simulated annealing and threshold-accepting algorithms are able to handle such time-dependent problems. Park (2000) presented the time-dependent VRP in which the travel speed between two locations depends on the route and the time of the day. They proposed a model for estimating the time varying travel speed. Ichoua, Gendreau, and Potvin (2003) proposed a time-dependent model for the VRPTW. The model that they developed is based on time-dependent travel speeds and satisfies the first-in-first-out (FIFO) property. They extended the tabu search heuristic to solve the problem and showed that the time-dependent model provides substantial improvements over a model based on fixed travel-times.

3. Quantification of quality loss

It is difficult to define the quality models for perishable foods since food degradation depends on various parameters and there exist many criteria on how to measure it.

Sloof, Tijskens, and Wilkinson (1996) presented a study on how to estimate the quality of the perishable products and use the effect-decomposition in the presented models. Verbič (2004) presented a study of the parameters affecting preservation of perishable goods in the cold chain. The studied parameters are the intensity of deterioration of goods at given levels of production. Bogataj, Bogataj, and Vodopivec (2005) studied the effects of perturbations in a supply chain (production or distribution part) on the stability of perishable goods in order to keep the product at the required level of quality and quantity for the final delivery. They present a model in the time domain and compare it with the formulation in the frequency space.

The above methods are very complex and require the estimation of many parameters which are not available in the distribution process. Therefore, we have extended the simple linear model proposed by Pawsey (1995) to estimate the decrease of fresh vegetables quality. We assume that any perishable product has a limited life-span under given conditions which is divisible into two stages. The point $t = 0$ represents the optimal condition of the vegetables; generally at the moment of harvest. During the first period of apparent quality stability (from 0 to A in Fig. 1), the quality is reducing but there are no discernible changes. At point A noticeable changes start in one or more of the quality parameters. During the second stage (from A to B in Fig. 1), the changes continue and at point B the product is unacceptable. In real-life the distribution is sel-

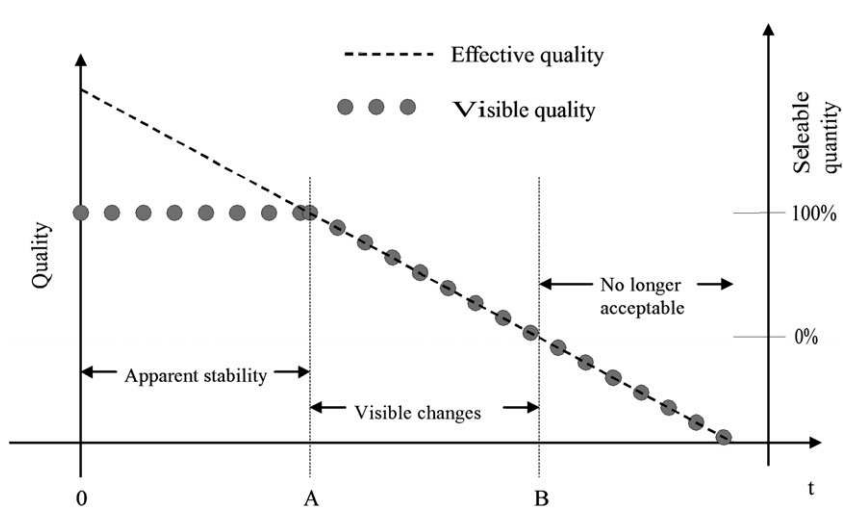

Fig. 1. Real and apparent quality loss with time.

dom completed in the 0–A period, and it is normal that the quality of the goods is reduced before delivery.

The time that elapses between harvesting and distribution is often quite high. In Slovenia, it is usually greater than 24 h, especially for imported vegetables. Most vegetables begin to visibly degrade after this period. Considering these data, we have made the assumption that distribution in Slovenia starts at point A.

Therefore the quality, Q [0,1], in the period A–B can be considered as a linear function as follows

$$Q = 1 - \frac{t - A}{B - A}.$$

We also assume that

$$Q \propto \text{Saleable Quantity} \propto \text{Commercial Value}.$$

A loss of quality of 20% can be related to either a load where 20% of transported quantity is completely damaged and 80% is in perfect conditions, a load where the whole transported quantity was evenly damaged so that it could be sold only at 80% of its original price, or a combination of these. Slovenian distribution companies and wholesalers estimate that the loss of quality of fresh vegetables during the transportation is on average between 8% and 23%. Based on these estimates we have assumed that the distribution schedule organized without considering the loss of quality in the objective function always ends before point B is reached.

It is difficult to define the actual rate of deterioration of the quality of a load over time, as it is dependent on the storage conditions and type of vegetable transported. However, it is not necessary to estimate the rate of deterioration to compare two different schedules of the same load. If we know the final quality of one of the schedules then we can estimate the relative final quality of the others.

To compare the loss of the quality of two different schedules we use the parameter Load×Time$_i$

$$\text{Load} \times \text{Time}_i = \sum_{j=0}^{N=i-1} \text{TravelTime}_{j(j+1)} \cdot C_{ji}, \tag{1}$$

where N_i, number of the customers for vehicle i; C_{ji}, load transported by the vehicle i on the path between the customer j and the customer $j+1$ (kg) and TravelTime$_{j(j+1)}$, time spent by vehicle i between the start of servicing of customer j and the start of the servicing of customer $(j+1)$ (min)

C_{ij} can be calculated using

$$C_{ji} = \sum_{k=j}^{N_i} c_k,$$

where c_k is the load to be delivered to customer k (kg).

Load×Time$_i$ represents the quantity in kg delivered by the vehicle i to its customers, multiplied by the time in minutes needed for the delivery. When the linear dependency of quality versus time is fixed, this quantity is proportional to the absolute loss in kg of saleable quantity. This means for example, that an increase of 20% of Load×Time causes a 20% reduction of saleable quantity. If we have two different schedules and know the loss of the quality of one, then we can estimate the effective loss of the second schedule using this parameter. The loss of quality represents an additional cost for the sellers (the complete chain of wholesalers, middlemen and shopkeepers in the distribution process) because a part of the purchased goods cannot be sold on the market or it can be sold, but not with the best market price.

Table 1
Financial parameters for the distribution of fresh vegetables on the Slovenian market

	Variable	Value
Sell price of vegetables	SellPrice	1.52 €1/kg
Operating cost of vehicles	PriceKm	0.6 €/km

The acquisition and the selling price of vegetables may vary during the year dependent on the vegetable type. We have calculated an average representative price for vegetables considering the typical quantities consumed of the different vegetables available (Vadnal & Bratuša, 2002) and considering the average prices on the Slovenian market (selling and acquisition price) over the period May–October 2005, reported in the Slovenian magazine Kmečki glas that weekly publishes the prices of the more important vegetables on the market. On the basis of these average prices we have calculated the "sell price" of the vegetables reported in Table 1.

4. Mathematical model for the transportation of the vegetables

We model the problem of the distribution of the vegetables between the central depot and the final customers (retailers) as a VRPTWTD. In our model we assume:

- homogeneous fleet of vehicles with a limited capacity;
- loss of the load quality during delivery as a linear function of time;
- time-dependent travel-times between the customers;
- soft time windows for all customers (time window may be violated at a penalty cost);
- hard time windows for the depot (violation of time window is not allowed).

In the transportation of vegetables in Slovenia, the time windows are usually only indicative. This and the fact that the travel-times fluctuate were the reasons to consider soft time windows. If $[a_i, b_i]$ represents the time window of customer i then, if the vehicle arrives at customer i before time a_i, it must wait up to time a_i before starting its service. If it arrives after b_i, a lateness penalty is incurred and is calculated using

$$\text{Penalty}_i = \begin{cases} (S_i^k - b_i) \cdot \text{Delayed Cost Factor}_i & S_i^k \geqslant b_i \\ 0 & S_i^k \leqslant b_i \end{cases} \tag{2}$$

where S_i^k, arrival time of vehicle k at customer i (min) and $\text{DelayedCostFactor}_i$, cost for a minute of lateness defined for customer i (€/min).

The objective function to minimize total cost is

$$\text{TotalCost} = k_1 \sum_{i=1}^{\text{NV}} \text{Distance}_i + k_2 \sum_{i=1}^{\text{NV}} \text{Time}_i + k_3 \sum_{i=1}^{\text{NV}} \text{Penalty}_i + k_4 \sum_{i=1}^{\text{NV}} \text{Load} \times \text{Time}_i, \tag{3}$$

where NV, number of vehicles; Distance_i, distance in km traveled by the vehicle i; Time_i, time in minutes that the vehicle i spend on the route; Penalty_i, time of delay in minutes that the vehicle i accumulate on the route; $\text{Load}\times\text{Time}_i$, loss of quality at vehicle i, calculated as in (1) and k_1, k_2, k_3 and k_4, parameters that represent the weight of each part of the total cost.

In the minimization of the objective function, it is not the value given to each k that is important, but the ratio between them. In the preliminary tests we have found that the most representative ratio between k_1, k_2 and k_3 for our problem is equal to 1. The units of the parameters are as follows: k_1 [€/km], k_2 [€/min], k_3 [–], k_4 [€/(min kg)].

By reducing the total $\text{Load}\times\text{Time}_i$ we reduce the time that the vegetables spend on the vehicles. This quantity does not take into account the time-traveled with empty vehicles; therefore, it is not the same as the Time_i quantity.

The travel-time between the customers is time-dependent, i.e., the travel-time between two locations depends on both the distance and on the time of day. Due to the time-dependency of the travel-times, the distance-traveled is also time-dependent (different routes for different times can be chosen in order to reduce the travel-time). To simplify the problem, we consider only four different time periods and suppose that the travel-time and the distance-traveled between the customers in each time period is constant (it is a step function of time).

To calculate the distance-traveled and the travel-times between the customers in each time period, we use the position of the customers. If the customers are defined on the real road network, we use the network characteristics and solve the time-dependent shortest path problem as in our previous work (Osvald, 2005). Otherwise, we use the corrected Euclidean distances.

To assure that the final solution of the VRPTWTD satisfies the FIFO property defined in Sung, Bell, Seong, and Park (2000) it is necessary to have continuous travel-times. Therefore, we round off travel-times in order to achieve time-continuous travel-times. An example of rounding of travel-times where four time periods during the day are used is presented in Fig. 2. If a vehicle starts

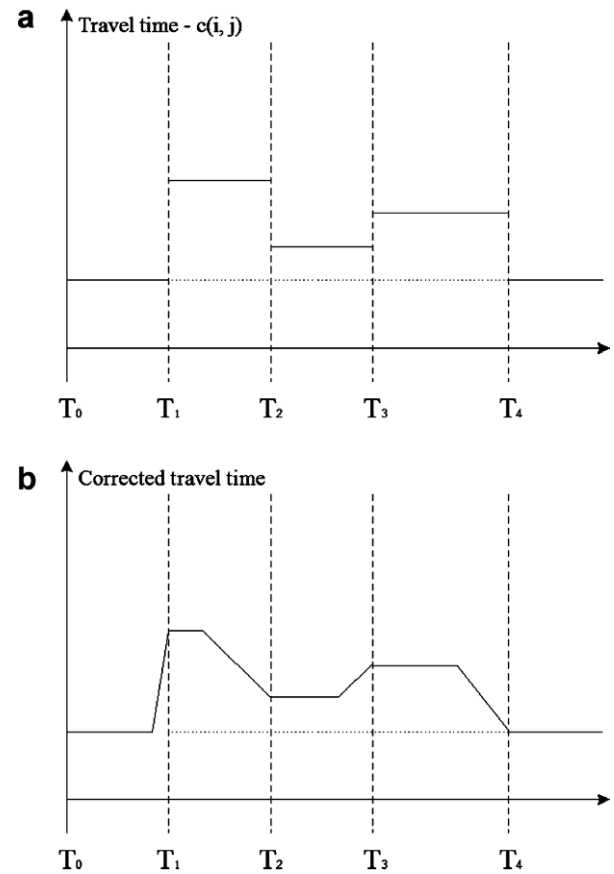

Fig. 2. Rounding of the travel-times for the four time periods used in the solution.

to travel on one arc near the end of one time period, then it cannot go the whole length of the arc before the end of the time period. The vehicle will therefore use two different speeds to cover the complete distance of the arc. If Time_A is the time necessary to go the whole length of the arc in the time period $[T_0, T_1]$ and Time_B is the time necessary to go the whole length of the arc in the time period $[T_1, T_2]$, then the rounding of the travel-time in the first and second time period is represented as follows

$$\text{TravelTime}\ (t) = \begin{cases} \text{Time}_\text{A} & t \leqslant T_1 - \text{Time}_\text{A}, \\ \text{Time}_\text{A} + \frac{\text{Time}_\text{B} - \text{Time}_\text{A}}{\text{Time}_\text{A}} \cdot (t - T_1 + \text{Time}_\text{A}) & T_1 - \text{Time}_\text{A} < t \leqslant T_1, \\ \text{Time}_\text{B} & T_1 < t \leqslant T_2 - \text{Time}_\text{B}. \end{cases}$$

Similar representations can be defined for other time periods.

5. Solution of the problem

To solve the problem, we have extended our previous method in Osvald (2005), developed to solve the VRPTWTD. The outline of the method used is

```
Repeat for different Start solutions (different Start Customers and different
  insertion costs)
     Find a feasible route:
       Solve classical VRPTW with an average travel-time and travel distance
       Calculate the effective arrival time using time-dependent travel-times
       For each not feasible route
           Repeat until route is not feasible
               Remove customer from the route
           End Repeat
       End For
       Insert the removed customers in other routes (if it is not possible,
         initialize new routes for the removed customers)
end Repeat
Select the three best routes
For each route
     Route reduction (reduce the number of routes)
     Insert stopover at the depot into each route where possible
     Insert free vehicles into the route
     Improve the solution with tabu search
end For
```

Initially a sequential constructive heuristic that solves a classical VRPTW is used; this approximates the travel-time and the travel distance between each pair of customers with the average travel-time and the average travel distance. Different initial solutions are calculated, since Bräysy (2001) showed that changing the initial customers and using different functions for the evaluation of the insertion costs of the customers (insertion cost represents the additional cost of the route when a new customers is inserted into it) yields much better solutions. The insertion cost is represented by a linear function which includes additional time and delay of the route, when a new customer is inserted. At each iteration (i.e., in each calculation of the initial solution), different coefficients for the additional time and additional delay are considered.

Then a correction procedure is used to take into account the actual time dependencies of the travel-times. In this procedure, the customers' delivery times are recalculated using time-dependent travel-times. When necessary, some customers are relocated in the route, or removed from it in order to make the route feasible. The removed customers are, when possible, reinserted in other routes. Otherwise, a new route or routes are initialized for the removed customers.

In the generation of the initial solutions we do not consider the loss of quality ($k_4 = 0$), since preliminary tests have shown that the consideration of these losses gives initial solutions with high lateness and high distances which cannot be always reduced in the following steps of the used method. After the generation of the initial solution we consider the objective function, including the loss of quality ($k_4 \neq 0$).

To further reduce the loss of saleable quantity it is necessary to reduce as far as possible the time that the vegetables spend on the vehicles. This could be achieved if vehicles perform shorter routes and return more often to the depot to take appropriately preserved vegetables. In this case the total traveled time would be higher, but the time that the vegetables spend on the vehicles is reduced. So, in the second part we perturbate the solution. First, we force the vehicles to go back to the depot in order to take appropriately preserved vegetables (at the depot, the vehicle has to wait for a specified time to complete loading operations). Therefore we introduce, where possible, new stopovers at the depot during the route with a defined stop time at the depot. The outline of the procedure is

```
Current location = last location
Repeat until Current location = 0
  Insert a ''virtual'' depot into the route
  at current location
  If the depot time window is not violated
      Current solution = new solution
  Current location = Current location-1
End repeat
```

Second, since it is not always possible to add new stopovers at the depot in the existing routes (indicated as 0′ in Fig. 3), we also add empty vehicles which have at least one additional stopover at the depot (only in this way we can be sure that the solution in which a vehicle stops at the depot, is taken into account in the optimization process). Usually we add two vehicles to each solution in order to extend the space of the possible solutions.

Sequence of customers (identified by numbers) before the insertion of the virtual depots (indicated with 0):

0 → 23 → 12 → 2 → 5 → 15 → 0
0 → 78 → 10 → 24 → 18 → 0
0 → 30 → 45 → 7 → 8 → 40 → 44 → 0

After the insertion of the virtual depots:

0 → 23 → 0' →12 → 0' → 2 → 5 → 15 → 0' → 0
0 → 0' → 78 → 10 → 24 → 18 → 0
0 → 30 → 0'→ 45 → 7 → 8 → 40 → 44 → 0 → 0
0 → 0' → 0
0 → 0' → 0

Fig. 3. Example of insertion of new stopovers at the depot and new empty vehicles.

The next step consists of applying the improvement heuristic based on the tabu search, which is a local-search method that uses memory structures. This method starts from an initial solution and at each iteration generates a neighborhood, i.e., a set of solutions around the current solution. The best solution in this neighborhood becomes the next current solution, even if it is worse than the current one. By further investigating the worse solutions, it is possible to avoid the trap of local optimum solutions. To generate a neighborhood we use a cross-exchange move in which two segments from two routes are taken and then swapped. An example is presented in Fig. 4. In the VRPTWTD it is not always possible to achieve a feasible solution starting from the infeasible one. So we consider only feasible solutions in the neighborhood generation.

The improvement heuristics is executed three times. The maximum number of iterations was fixed at 2000. To develop the improvement heuristic based on the tabu search we have used EasyLocal++, an object-oriented framework for the design and analysis of local-search algorithms, as presented in Di Gaspero and Schaerf (2001).

Osvald (2005) provides detailed presentation of the neighborhood generation (including the software code) and the calculations of other parameters (e.g., the length of the tabu list, maximum number of iterations). In the improvement heuristic most of the additional stopovers at the depot and the additional vehicles are subsequently removed in order to reduce the total distribution cost.

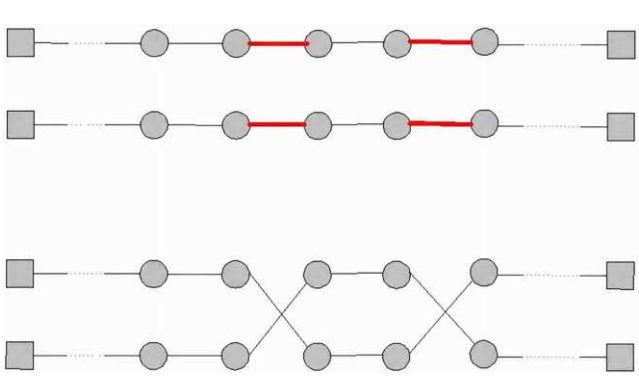

Fig. 4. Example of a cross-exchange move.

However, the remaining additional stopovers and vehicles generate a quality saving greater than their additional cost.

6. Performance of the solution method

6.1. Definition of the data set

Ideally the method would be tested on real data, considering real customers and traffic conditions. Unfortunately, the wholesalers operating in Slovenia would not make such commercially sensitive data such as the effective arrival time at the customers, the exact travel-time that the vehicles spend on the route and the exact delivered order quantity available in the public domain. We know the total distributed quantity of vegetables for different areas, but not individual customer orders. Additionally, we do not know the exact location of all customers and acquisition of this data would require a great effort including the collection of all addresses and their mapping on a vector map.

Therefore, we decided to use some representative data sets in which the distribution of the customers and requests are known. We chose the classical Solomon's instances (Solomon, 1987). Each of the 56 instance has 100 customers generated within a 100 by 100 km square area. The distances between the customers are Euclidean. The problems are partitioned into six different sets, namely C1, C2, R1, R2, RC1 and RC2. The customers are uniformly distributed in the problems of type R, clustered in groups in the problems of type C and mixed in the problems of type RC. In the problems of type 1, a small number of customers can be serviced on each route due to a narrow time window at the depot, as opposed to problems of type 2 where each route may have many customers.

These data sets are sufficient to test the validity of the solutions since they are constructed so that they each consider a different distribution of the customers. Also, the best solutions for number of vehicles used, total distance-traveled and the total travel-time are known for these problems so we can compare these reference case solutions to solutions if the consideration of load quality is added.

In order to achieve representative data for the Slovenian market, we have analyzed various independent wholesalers that operate in Slovenia. The largest has three distribution centers and about 700 retailers dispersed across Slovenia. The other wholesalers are smaller and usually operate within a limited footprint, not covering the whole distribution area. The most common vehicle in use is not insulated, has a capacity of 10,000 kg, and the operating cost is about 0.6 €/km, so this was used in the data sets.

6.2. Definition of the tests

To evaluate the performances of the proposed optimization method we performed the following tests:

- A1: Solution of the classical Solomon's instances using the method presented by Osvald (2005) with $k_1 = k_2 = k_3 = 1$ and $k_4 = 0$ in Eq. (3) (i.e., we do not consider the loss of quality).
- A2: Solution of the classical Solomon's instances using the method presented by Osvald (2005) with $k_1 = k_2 = k_3 = k_4 = 1$ in Eq. (3) (i.e., consideration of loss of quality).
- A3: Solution of the classical Solomon's instances using the method presented in Section 5 with $k_1 = k_2 = k_3 = k_4 = 1$ in Eq. (3).
- A4–A6: Same as in A1–A3, respectively, but using modified Solomon's instances considering time-dependent travel-times.

To take into account the time-dependent travel-times, we consider four time periods and generate four different matrices, multiplying the origin destination (OD) matrices in the Solomon instances with four different correction factors (namely 0.8, 1.2, 0.85 and 1.15). Each matrix represents the travel-time in one of the four time periods. We define the time periods so that for each instance the average travel-times between the customers are as the same as those in the Solomon's instance. Finally we rounded the travel-times to give a continuous travel-time between the customers as shown in Fig. 2.

In tests A3 and A6 we set stops at the depot to be three times the maximum time necessary to unload the order at the customers. This upper-limit was defined, since no exact information about the real time required at the depot was available.

The total transported quantity (TotLoad) for each set of problems was calculated as follows

$$\mathrm{TotLoad} = \mathrm{TotWC} \cdot \mathrm{NV}. \tag{4}$$

NV is the number of vehicles used in test A1 and reported in Table 2 (e.g., 11.08 vehicles used for R1, 10 used vehicles for C1). TotWC is the average load weight of each vehicle. The weight capacity of the vehicles is usually not completely used due to restrictions such as volume of the load, the distribution of the customers across the delivery area and time windows. To quantify the effective use of the capacity, we calculated

$$\mathrm{WR} = \frac{\mathrm{TotalTransportedQuantity}}{\mathrm{VehicleCapacity} \cdot \mathrm{NV}},$$

where the TotalTransportedQuantity and the VehiclesCapacity is expressed in the units defined in the Solomon's instances. Further we restricted vehicle weight capacity to 80% of the nominal value. Therefore, the total weight capacity can be calculated as

$$\mathrm{TotWC} = 0.8 \cdot \mathrm{WR} \cdot 10{,}000\ \mathrm{kg}.$$

7. Results and discussion

Tables 2–7 present the results for tests A1–A6. In these tables, for each set of Solomon's instances, we report:

- Solomon's class: name of the set of Solomon's instances. The results reported in the tables are the average value of all instances of the set.
- Number of vehicles: the average of the number of vehicles used for all instances of the set.
- Distance: the average of the total distances traveled in km for all instances of the set.
- Time: the average of the total traveled times in minutes for all instances of the set.
- Lateness: the average of the total lateness in minutes for all instances of the set.
- Load×Time: the average of the loss of quantity for all instances of the set calculated using Eq. (1).
- Loss reduction: the percentage reduction of Load×Time in the tests where we consider loss of quality in the objective function compared to tests that did not consider loss of quality.
- Additional profit: the profit that could be achieved, when due to the reduction of the losses, more vegetables can be sold to the customer; it is calculated using

$$\begin{aligned}\mathrm{AdditionalProfit}_{Ai} = {} & (\text{Loss Reduction } \mathrm{i}_{Ai}) \cdot \frac{\mathrm{EffectiveLoss}}{100} \\ & \times \mathrm{TotLoad} \cdot \mathrm{SellPrice} \\ & + (\mathrm{TotalDistance}_{Ai} - \mathrm{TotalDistance}_{A1}) \cdot \mathrm{PriceKm}\end{aligned} \tag{5}$$

where EffectiveLoss, the percentage loss of saleable quantity in the schedules (A1 or A4) and TotLoad, the total transported quantity or the total demand in kg calculated using Eq. (4). SellPrice and PriceKm are defined in Table 1.

Table 3 shows the results for the test A2 in which we have considered the loss of quality. Here we have an

Table 2
Results for Solomon's instances, not considering the loss of quality

Solomon's class	Number of vehicles	Distance (km)	Time (min)	Lateness (min)	Load×Time (min kg)	WR	Total load
R1	11.08	1212.6	2270.6	106.8	115406.2	0.65	57616
Cl	10.00	844.2	9833.6	0.0	601233.3	0.90	72000
RC1	11.50	1375.3	1429.9	60.4	143593.8	0.75	69000
R2	2.91	1191.6	2352.7	43.7	608945.6	0.50	11640
C2	3.00	632.9	9637.9	0.0	534355.0	0.86	20640
RC2	3.25	1390.9	2679.3	93.5	672464.5	0.53	13780

Table 3
Results for Solomon's instances considering the loss of quality using the solution method presented by Osvald (2005)

Solomon's class	Number of vehicles	Distance (km)	Time (min)	Lateness (min)	Load×Time (min kg)	Loss reduction (%)	Additional profit (€) – effective loss = 8	Additional profit (€) – effective loss = 23
R1	11.8	1432.3	2299.5	650.0	89285.9	22.6	1453.9	4427.1
Cl	10.0	1131.9	10295.0	1509.2	551480.1	8.3	350.7	1331.8
RC1	12.0	1564.3	2610.4	504.3	119361.5	16.9	1113.7	3414.6
R2	3.0	1538.6	2731.3	1149.3	490155.9	19.5	150.8	823.8
C2	3.0	916.4	9912.5	4800.3	404297.6	24.3	291.6	1157.3
RC2	3.5	1772.5	2951.8	1463.1	604109.1	10.2	−20.1	371.6
Average results						17.0	556.8	1921.0
Average results for R1, Cl and RC1						15.9	972.8	3057.8

Table 4
Results for Solomon's instances using the method presented in Section 5

Solomon's class	Number of vehicles	Distance (km)	Time (min)	Lateness (min)	Load×Time (min kg)	Loss reduction (%)	Additional profit (€) – effective loss = 8	Additional profit (€) – effective loss = 23
R1	12.5	1532.2	2656.0	362.1	80825.0	30.0	1907.6	5843.9
C1	12.0	2737.8	14179.2	5505.4	417769.0	30.5	793.4	4411.2
C1	13.0	1688.5	2808.3	392.9	110199.8	23.3	1503.1	4673.9
R2	5.0	1986.2	3687.0	838.4	143019.0	76.5	931.2	3571.0
C2	5.0	2460.8	13430.5	56090.1	5246.6	64.2	121.7	2406.3
RC2	5.3	2162.8	3687.0	1744.4	198043.3	70.5	986.7	3705.1
Average results						49.2	1040.6	4101.9
Average results for R1, Cl and RC1						27.9	1401.4	4976.3

Table 5
Results for modified Solomon's instances with time-dependent travel-times, not considering the loss of quality

Solomon's class	Number of vehicles	Distance (km)	Time (min)	Lateness (min)	Load×Time (min kg)
R1	10.5	954.6	2091.0	81.3	137040.5
Cl	10.0	740.7	9793.0	9.0	745600.8
RC1	10.5	105.9	2206.9	77.1	178371.0

Table 6
Results for modified Solomon's instances with time-dependent travel-times considering the loss of quality using the solution method presented by Osvald (2005)

Solomon's class	Number of vehicles	Distance (km)	Time (min)	Lateness (min)	Load×Time (min kg)	Loss reduction (%)	Additional profit (€) – effective loss = 8	Additional profit (€) – effective loss = 23
R1	11.6	1127.8	2295.0	300.8	85637.9	37.5	2524.0	7451.4
Cl	10.0	890.0	10009.3	925.4	554137.7	25.7	1534.1	4578.6
RC1	12.1	1266.6	2428.1	177.0	114442.9	35.8	2497.7	7384.3
Average results						33.0	2185.3	6471.4

Table 7
Results for modified Solomon's instances with time-dependent travel-times using the method presented in Section 5

Solomon's class	Number of vehicles	Distance (km)	Time (min)	Lateness (min)	Load×Time (min kg)	Loss reduction (%)	Additional profit (€) – effective loss = 8	Additional profit (€) – effective loss = 23
R1	11.4	1097.3	2285.8	377.9	85404.8	37.7	2554.2	7503.9
Cl	10.0	865.5	9058.6	1876.4	95388.8	47.0	2895.1	8463.9
RC1	11.8	1217.6	2362.3	188.5	115086	.335.5	2500.9	7338.3
Average results						40.0	2650.1	7768.7

increase of vehicles used of up to 7.7%, an increase of distance-traveled between 13% and 48% and an average reduction of loss of quality equal to 17%. To quantify the improvement of the achieved solution, we have calculated the additional available profits (additional profit) considering two different percentages of loss of saleable quantity in the schedule A1 (effective loss = 8 and effective loss = 23 which represent the loss recognized by law and the loss occasionally experienced). The calculated quantities are significant; they show that the reduction of the loss of quality offers significantly higher savings than the increased cost due to the increased distance-traveled.

Of course, in real-life this additional profit cannot be always achieved because the demand on the market is limited, but our results clearly indicate the value that these losses of saleable quantities of vegetables represent on the market. Since this distribution activity is undertaken many times per weeks, the compound impact of these additional profit achieved across the year could be significant.

One of the parameters that it is also necessary to take into account is the total lateness. Even though we have a significant increase of the total lateness (from 0 to 4800 min for the problem C2), it remains acceptable in all instances except C2. In reality, a total lateness of 1000 min represents an average delay per customer of 10 min which is probably acceptable.

Table 5 shows the results for test A4. The Load×Time represents the average of the loss of quantity for the scheduling achieved in test A1, considering time-dependent travel-times (we consider the same customer sequence as in test A1, but we use the time-dependent travel-times to calculate the arrival time at the customer). The reduction of the loss of quality that we achieve considering time-dependent travel-times is nearly 20%. In reality, the travel-time plays an important role in the distribution of the perishable goods, since its fluctuations may extend the time that the goods remain on the vehicles and the resultant effect on the total loss of quality of the load.

By considering the loss of quality in the solution of the problem (test A4), we achieve a significant reduction of the loss of quality as presented in Table 5.

Tables 4 and 7 present the results for tests A3 and A6 in which the method presented in Section 5 was used. In test A3, the increase of vehicle numbers is more significant than for A2 as is the improvement in the reduction in total losses. The additional profit increased nearly two times in comparison to test A2 for the average results, which demonstrates the advantages in the proposed method. The lateness is higher for classes C1 and C2 (not always acceptable in real-life) and acceptable in all other classes.

In A6 (Table 7) we have an average improvement of 7% on the total loss of quality compared to test A5. Moreover, there is a reduction of vehicle numbers and a reduction of the distance-traveled. The introduction of new stopovers (presented in Section 5) allows, in some cases, the generation of a good starting solution for the tabu search which can generate a final solution with a very low number of vehicles and a short distance traveled. The additional profits are higher than those achieved in the time-independent test A3.

In our method we have considered an upper-limit for the time at the depot. A reduction of the stop time at the depot improves the reduction of the loss of the quality, since more stopovers at the customers are possible and so the time that the vegetables pass on the vehicle is reduced.

8. Conclusions

The efficiency of the distribution process of perishable foods, where time the food spends on the vehicles is the most significant and crucial factor, represents an important problem in food distribution. For a distribution network based on the Slovenian vegetable market, for constant travel-times the proposed method gave solutions with 27.9% average extra reductions in the loss of quality relative to the solution based on the model in which the loss of quality was not taken into account. The savings increased on average to 40% when time-dependent travel-times were taken into account.

References

Amponsah, S. K., & Salhi, S. (2004). The investigation of a class of capacitated arc routing problems: The collection of garbage in developing countries. *Waste Management, 24*, 711–721.

Bentner, J., Bauer, G., Obermair, G. M., & Morgenstern, I. (2001). Optimization of the time-dependent traveling salesman problem with Monte Carlo methods. *Physical Review E*, 036701-1–036701-8.

Bogataj, M., Bogataj, L., & Vodopivec, R. (2005). Stability of perishable goods in cold logistic chains. *International Journal of Production Economics, 93–94*(1), 345–356.

Bräysy, O. (2001), Efficient local search algorithms for the vehicle routing problem with time windows. Working Paper, *SINTEF Applied Mathematics*, Department of Optimization.

Di Gaspero, L., & Schaerf, A. (2001). Writing local search algorithms using EASYLOCAL++. In S. Voß & D. L. Woodruff (Eds.).

Optimization software class libraries, OR/CS. Boston: Kluwer Academic Publishers.

Faulin, J. (2003). Applying MIXALG procedure in a routing problem to optimize food product delivery. *Omega, 31*(5), 387–395.

Fu, L. (2002). Scheduling dial-a-ride paratransit under time-varying, stochastic congestion. *Transportation Research Part B, 36*(6), 485–506.

Gendreau, M., Laporte, G., & Seguin, R. (1996). Stochastic vehicle routing. *European Journal of Operational Research, 88*(1), 3–12.

Hadjiconstantinou, E., & Roberts, D. (1998). A computational approach to the vehicle routing problem with stochastic demands. In P. Borne, M. Ksouri, & A. El Kamel (Eds.), *Computational engineering in systems applications: 16th European conference on operational research (EURO XVI)* (pp. 139–144). Brussels, Belgium: IEEE.

Hwang, H. S. (1999). A food distribution model for famine relief. *Computers and Industrial Engineering, 37*(1–2), 335–338.

Ichoua, S., Gendreau, M., & Potvin, J. Y. (2003). Vehicle dispatching with time-dependent travel times. *European Journal of Operational Research, 144*(2), 379–396.

Kenyon, A., & Morton, D. P. (2003). Stochastic vehicle routing with random travel times. *Transportation Science, 37*(1), 69–82.

Knavs, S. (2003). *Ocena učinkovtosti tržnih poti za zelenjavo v Sloveniji*. Thesis research, Biotehniska fakulteta, Univerza v Ljubljani, Ljubljana.

Križaj, J. (2002). *Mednarodni cestni prevoz zelenjave*. Thesis research, Biotehniska fakulteta, Univerza v Ljubljani, Ljubljana.

Malandraki, C., & Daskin, M. S. (1992). Time dependent vehicle routing problems: Formulations, properties and heuristic algorithms. *Transportation Science, 26*(3), 185–199.

Osvald, A. (2005). *The vehicle routing problem with time windows and time dependent travel times: A heuristic approach*. Ph.D. thesis, Università degli studi di Trieste, Trieste.

Park, Y. B. (2000). A solution of the bicriteria vehicle scheduling problems with time and area-dependent travel speeds. *Computers and Industrial Engineering, 38*(1), 173–187.

Pawsey, R. K. (1995). Preventing losses and preserving quality in food cargoes. In J. L. Albert (Ed.). *Food, nutrition and agriculture – 15 – food safety and trade*. Italy: Food and Agriculture Organization (FAO) of the United Nations.

Prindezis, N., Kiranoudis, C. T., & Marinos-Kouris, D. (2003). A business-to-business fleet management service provider for central food market enterprises. *Journal of Food Engineering, 60*(2), 203–210.

Sloof, M., Tijskens, L. M. M., & Wilkinson, E. C. (1996). Concepts for modeling the quality of perishable products. *Trends in Food Science and Technology, 7*, 165–171.

Solomon, M. M. (1987). Algorithms for the vehicle routing and scheduling problem with time window constraints. *Operations Research, 35*, 254–265.

Sung, K., Bell, M. G. H., Seong, M., & Park, S. (2000). Shortest paths in a network with time-dependent flow speeds. *European Journal of Operational Research, 121*(12), 32–39.

Tarantilis, C. D., & Kiranoudis, C. T. (2001). A meta-heuristic algorithm for the efficient distribution of perishable foods. *Journal of Food Engineering, 50*(1), 1–9.

Tarantilis, C. D., & Kiranoudis, C. T. (2002). Distribution of fresh meat. *Journal of Food Engineering, 51*(1), 85–91.

Toth, P., & Vigo, D. (2002). *The vehicle routing problem. Monographs on discrete mathematics and applications*. Philadelphia: SIAM.

Vadnal, K., Bratuša, A. (2002). Tržne poti slovenskih ekoloških kmetij. In: *Novi izzivi v poljedelstvu* (pp. 284–288). Ljubljana.

Verbič, M. (2004). *Econometric estimation of parameters of preservation of perishable goods in cold logistic chains*. Working Paper No. 24, Institute for Economic Research, Ljubljana.